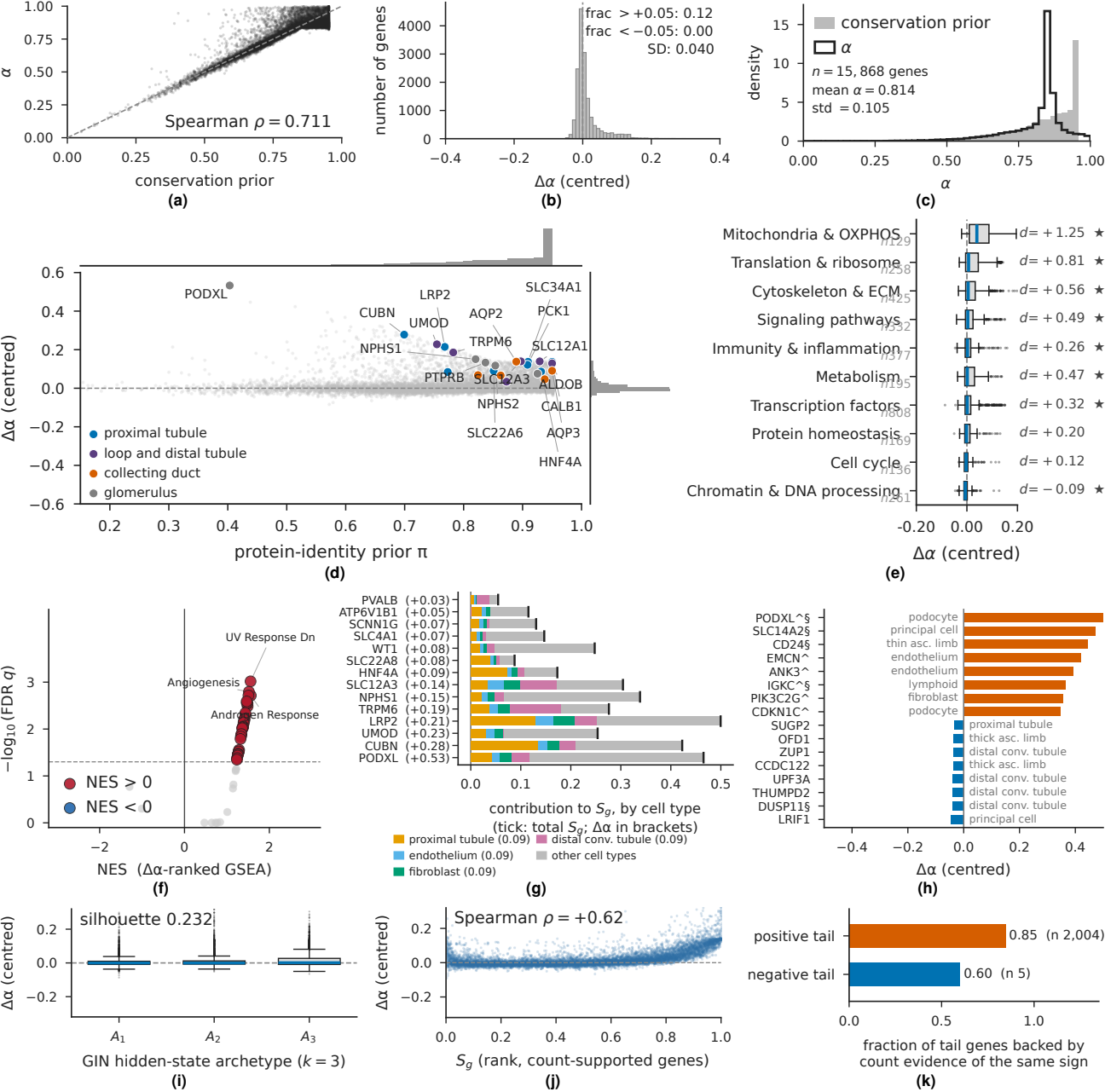

Spearman ρ = 0.711
conservation prior
α
(a)
frac > +0.05: 0.12
frac < −0.05: 0.00
SD: 0.040
number of genes
Δα (centred)
(b)
conservation prior
α
n = 15,868 genes
mean α = 0.814
std = 0.105
density
(c)
PODXL
LRP2
SLC34A1
CUBN
UMOD
AQP2
PCK1
NPHS1
TRPM6
SLC12A1
PTPRB
SLC12A3
ALDOB
NPHS2
CALB1
SLC22A6
AQP3
HNF4A
proximal tubule
loop and distal tubule
collecting duct
glomerulus
Δα (centred)
protein-identity prior π
(d)
Mitochondria & OXPHOS n129 d = +1.25 ★
Translation & ribosome n258 d = +0.81 ★
Cytoskeleton & ECM n425 d = +0.56 ★
Signaling pathways n932 d = +0.49 ★
Immunity & inflammation n377 d = +0.26 ★
Metabolism n195 d = +0.47 ★
Transcription factors n808 d = +0.32 ★
Protein homeostasis n169 d = +0.20
Cell cycle n136 d = +0.12
Chromatin & DNA processing n261 d = −0.09 ★
Δα (centred)
(e)
UV Response Dn
Angiogenesis
Androgen Response
NES > 0
NES < 0
−log10(FDR q)
NES (Δα-ranked GSEA)
(f)
PVALB (+0.03)
ATP6V1B1 (+0.05)
SCNN1G (+0.07)
SLC4A1 (+0.07)
WT1 (+0.08)
SLC22A8 (+0.08)
HNF4A (+0.09)
SLC12A3 (+0.14)
NPHS1 (+0.15)
TRPM6 (+0.19)
LRP2 (+0.21)
UMOD (+0.23)
CUBN (+0.28)
PODXL (+0.53)
contribution to S_g, by cell type
(tick: total S_g; Δα in brackets)
proximal tubule (0.09)
distal conv. tubule (0.09)
endothelium (0.09)
other cell types
fibroblast (0.09)
(g)
PODXL^§ podocyte
SLC14A2§ principal cell
CD24§ thin asc. limb
EMCN^ endothelium
ANK3^ endothelium
IGKC^§ lymphoid
PIK3C2G^ fibroblast
CDKN1C^ podocyte
SUGP2 proximal tubule
OFD1 thick asc. limb
ZUP1 distal conv. tubule
CCDC122 thick asc. limb
UPF3A distal conv. tubule
THUMPD2 distal conv. tubule
DUSP11§ distal conv. tubule
LRIF1 principal cell
Δα (centred)
(h)
silhouette 0.232
Δα (centred)
A_1
A_2
A_3
GIN hidden-state archetype ($k = 3$)
(i)
Spearman ρ = +0.62
Δα (centred)
S_g (rank, count-supported genes)
(j)
positive tail
0.85 (n 2,004)
negative tail
0.60 (n 5)
fraction of tail genes backed by count evidence of the same sign
(k)